Joint Resource Management and Load Balancing in Multi-Satellite Beam Hopping With Interference Suppression: An Energy Minimization Perspective

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***Abstract*—Low Earth orbit (LEO) satellite communications have received significant attention due to the global coverage and low-latency characteristics. Due to the limited size, weight, and power of LEO satellites, meeting the demand of satellite-terrestrial communications with minimal energy consumption while suppressing interference to co-existing systems remains a significant challenge. In this paper, a LEO satellite architecture with beam hopping (BH) is proposed, where the dynamic resource management and real-time load balancing are jointly implemented. Specifically, an optimization problem for the BH pattern, resource management, and load balancing with multi-satellite collaboration is formulated to achieve the long-term average energy minimization, while suppressing the interference to the co-existing system. Subsequently, by introducing the Lyapunov optimization method, the long-term energy minimization is transformed into a series of dynamic optimization problems for each time slot, thereby ensuring both the energy minimization and queue stability, where the block coordinate descent algorithm is employed to alternately address these issues. In particular, the BH pattern is determined by the multiplier penalty method combined with the majorization-minimization algorithm. Additionally, the resource management is addressed through the successive convex approximation technique, while real-time load balancing is achieved via quadratic programming. Overall, the proposed method, which can be effectively implemented within polynomial time, outperforms other benchmarks in simulations by achieving interference suppression in co-existing systems and reducing the energy consumption, while satisfying the communication demands.**

***Index Terms*—Satellite communications, beam hopping, multi-satellite collaboration, energy minimization, resource management, real-time load balancing, queue stability, interference suppression, Lyapunov optimization.**

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I. Introduction

IN RECENT years, mega-constellations of low Earth orbit (LEO) satellites, such as Starlink and Kuiper, have been widely deployed for communications [2], [3]. Compared to traditional satellite communications, the LEO satellite communication offers lower propagation delays and higher data rates [4], [5], making it a key technology for next-generation wireless communications [6]. However, LEO communication systems face significant challenges in resource management and energy optimization due to their high mobility, limited resources, and spectrum overlap with co-existing systems [7]. Consequently, enhancing the reliability and efficiency of LEO satellite communications while mitigating the interference has become a critical issue.

To overcome this challenge, multi-beam satellites have been employed to enhance the efficiency and data rates by dividing coverage areas into multiple cells, each served by a dedicated beam [8], [9]. Additionally, resource management and precoding among beams notably improved the performance of satellite-terrestrial communications [10], [11]. However, due to size, weight, and power (SWaP) constraints, only a limited number of beams can be equipped on a LEO satellite, and its swift motion introduces time-varying characteristics [12]. Therefore, beam hopping (BH) is adopted, where beams serve different cells in a time-division manner. Moreover, it enhances the flexibility in satellite communications and accommodates both the time-varying characteristics and the non-uniform distribution of communication demands. Consequently, BH has been recognized as an effective solution for LEO satellites and it has been discussed in 3GPP TR 38.811 [13].

Meanwhile, there have been several studies on satellite BH. Specifically, full-digital beamforming was applied to satellite beams, and a random search method was employed to determine the BH pattern in [14]. Besides, a BH pattern scheme that

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considered both communication demands and adjacent-beam interference avoidance was presented in [15]. In addition, literature [16] introduced a matching-theoretic approach to jointly optimize the BH pattern and resource allocation. However, these methods are based on pre-scheduling BH time plans, which are inadequate in time-varying environments [17]. As a result, the BH pattern scheduling based on user service weight gain (USWG) within each time slot was proposed in [18]. Meanwhile, deep reinforcement learning (DRL) was adopted in [19] to dynamically select BH patterns. Nevertheless, these works predominantly focused on single-satellite systems. In multi-satellite scenarios, the complexity of managing inter-satellite interference and the BH pattern increases rapidly. To address this, literature [20] utilized a terrestrial control center to pre-schedule multi-satellite BH time plans, while multi-agent DRL was applied for dynamic BH pattern selection in [21] and [22].

Moreover, due to the overlapping spectrum, the co-frequency interference between LEO satellites and co-existing systems should be considered. To address this, the interference between LEO satellites and geosynchronous orbit (GSO) satellite ground stations was avoided by pre-matching LEO satellites with terrestrial cells [7], [23]. Additionally, spectrum allocation was further introduced to suppress interference in [24]. However, these techniques still depend on pre-scheduling to realize interference avoidance with co-existing systems, and therefore lack adaptability to time-varying characteristics.

Besides, it has been verified that the dynamic resource management based on real-time channel states and queue lengths can effectively improve the communication performance in time-varying scenarios [25]. Specifically, the optimal cross-layer scheduling policy with hard delay constraints in the high signal-to-noise ratio regime was analyzed in [26]. Moreover, an optimal joint delay and energy control strategy was derived with a constrained Markov decision process framework [25]. In massive access systems, the mean-field approximation was adopted to reduce the computational complexity [27]. However, these approaches typically rely on the prior knowledge on channel state and introduce significant computational complexity. Consequently, the Lyapunov optimization has been demonstrated as an alternative method to achieve the near-optimal performance [28], and has been applied in pilot scheduling for massive machine-type communications [29], decentralized dynamic offloading in mobile edge computing [30], [31], and age-of-information optimization in downlink systems [32]. In satellite communications, dynamic resource management was investigated for a single multi-beam satellite [33].

Recently, the superiority of integrating load balancing with resource management has been investigated [34]. In multi-satellite communications, the non-uniform distribution of communication demand and varying interference levels to co-existing systems further highlight the necessity for load balancing [20], [23]. Existing approaches achieved load balancing by fixed pre-matching of LEO satellites to terrestrial cells based on the communication demand and interference level of each cell [7], [21]. Moreover, advanced laser inter-satellite links (ISLs) enable high data rate and long-distance communications, enhancing the feasibility of multi-satellite collaboration, which has been applied by satellite-based federated learning [35] and cooperative offloading [36]. Therefore, real-time load balancing by multi-satellite is not only feasible but also crucial for improving the communication performance.

Although significant progress has been made by the above literature, there are still several shortcomings. Firstly, few of these studies addressed the issue of minimizing energy consumption while satisfying the communication demands, which is essential for LEO satellites with limited SWaP and time-varying characteristics. Secondly, dynamic resource management in the case of multi-satellite BH communications with interference has not yet been investigated. In addition, existing load balancing strategies rely on fixed satellite-to-cell pre-matching, which lacks adaptability to dynamic conditions, and the potential of laser ISLs is not fully exploited. To address these issues, we make the following contributions:

1) A multi-satellite BH architecture for LEO satellite communication is proposed to minimize the long-term energy consumption and mitigate the interference to co-existing systems. By the Lyapunov drift-plus-penalty method, the long-term energy minimization is transformed into a series of optimization problems for each time slot with theoretical performance guarantees. Through joint resource management and load balancing in this multi-satellite BH architecture, both the energy minimization and queue stability are considered, meeting satellite communication demands and suppressing the interference to co-existing systems.
2) The BH pattern and resource management are optimized to enhance the communication performance. Specifically, to address the discrete variables and the computational complexity, the BH pattern is determined by a combination of the multiplier penalty (MP) method and the majorization-minimization (MM) algorithm, i.e., the MPMM approach. Moreover, the successive convex approximation (SCA) technique is applied to handle the non-convex constraints in resource management. These optimizations not only yield a significant reduction in energy consumption but also suppress interference to co-existing systems through the joint optimization of power and frequency segments.
3) Real-time load balancing via multi-satellite collaboration is implemented. Through multi-satellite collaboration, data packets are transferred via laser ISLs to satellites with more available resources and lower interference levels to GSO ground stations, thereby reducing the long-term average energy consumption. Additionally, quadratic programming is applied to obtain the load balancing scheme, and the extra energy overhead of laser ISLs during load balancing is also considered. Simulation results further demonstrate that jointly optimizing the BH pattern, resource management, and load balancing through the block coordinate descent (BCD) algorithm can effectively decrease the energy consumption and outperform other benchmarks.

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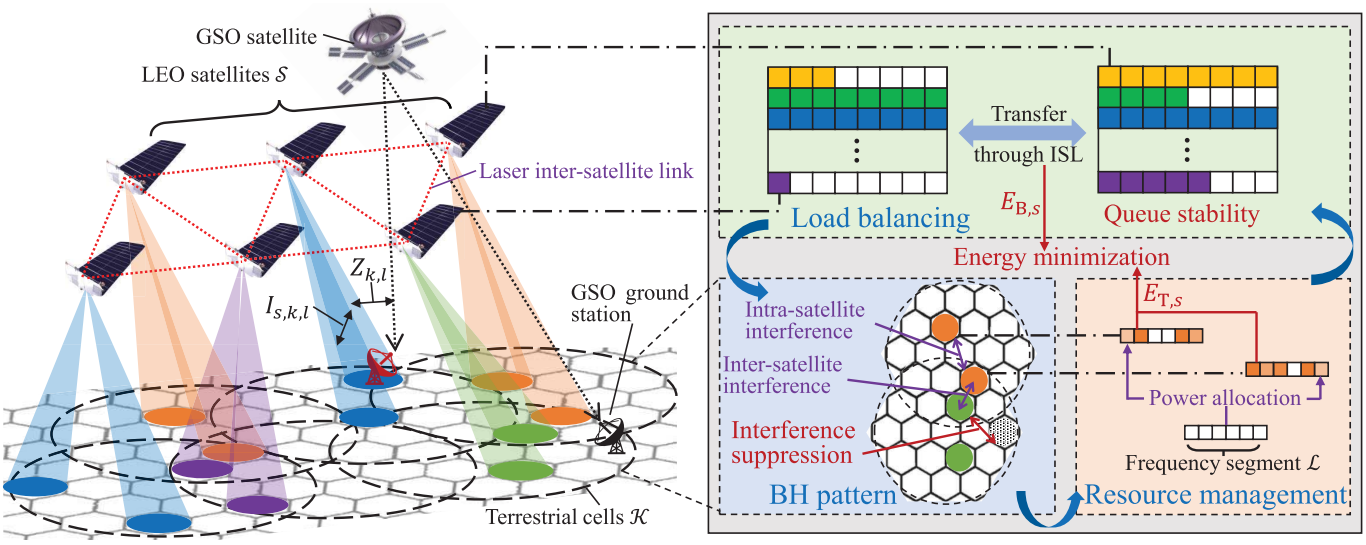

Fig. 1. The architecture of multi-satellite BH communications for energy minimization considering interference suppression and queue stability.

The paper is structured as follows. The system model for multi-satellite BH communications is introduced in Section II, encompassing the satellite network architecture, link models for LEO satellites to terrestrial cells and to GSO ground stations, and the queue model. In Section III, the optimization problem for minimizing energy consumption is formulated and transformed into a series of problems within each time slot by the Lyapunov drift-plus-penalty method. Moreover, the BCD algorithm is adopted in Section IV to alternately optimize the BH pattern, resource management, and load balancing. Besides, simulation results are illustrated and analyzed in Section V, followed by the conclusion in Section VI.

II. System Model

The architecture of the LEO satellite network is introduced in Section II-A. Moreover, the link and queue models are also established in Section II-B and Section II-C, respectively. For ease of reference, the primary notations and their definitions are summarized in Table I, and the index $[n]$ indicates that the variable is associated with the n-th time slot.

A. Satellite Network

As shown in Fig. 1, the downlink system from multiple LEO satellites to terrestrial cells is investigated in this paper. Moreover, to enhance the resource management flexibility and mitigate the co-frequency interference, the available spectrum resources in LEO satellite communications are divided into multiple frequency segments, each with a bandwidth of W, and the set of the frequency segment indices is denoted as $\mathcal{L}$. For convenience, the indices of all LEO satellites and all cells form the sets $\mathcal{S}$ and $\mathcal{K}$, respectively. Moreover, $|\mathcal{S}|$ represents the number of elements in set $\mathcal{S}$, corresponding to the number of the LEO satellites, and similarly the numbers of cells and frequency segments are denoted by $|\mathcal{K}|$ and $|\mathcal{L}|$.

Additionally, each LEO satellite has N_b beams to serve its coverage area, with the cells covered by LEO satellite s forming set $\Omega(s)$. Correspondingly, each cell may fall within the coverage of multiple LEO satellites, and the set of LEO satellites that can serve the k-th cell is denoted as $\Phi(k)$. Meanwhile, ground stations associated with the GSO satellites also exist in some cells, which form the subset $\mathcal{K}_{\mathrm{G}} \subset \mathcal{K}$. Besides, the spectrum of the GSO ground stations may overlap with LEO satellite communications. Specifically, for the ground station in the k-th cell where $k \in \mathcal{K}_{\mathrm{G}}$, the set of overlapping frequency segments between LEO and GSO satellites is denoted as $\mathcal{L}_k$.

Generally, an efficient scheduling policy for beams, resources, and loads is essential in satellite-terrestrial communications due to the limited number of beams, non-uniform temporal and geographical distribution of communication demands, and the interference suppression for co-existing systems. Therefore, beams are scheduled in a time-division manner to illuminate different cells across time slots, referred to as BH, while the duration of each time slot is T_0, and its index set is denoted as $\mathcal{N}$. Besides, the dynamic resource management is executed among time slots, beams, and frequency segments to reduce the co-frequency interference and suppress interference to GSO ground stations. In addition, laser ISLs enable both inter-satellite data transmission and information exchange, facilitating load balancing and multi-satellite collaboration. Consequently, through multi-satellite collaboration, the BH pattern, resource management, and load balancing are determined in each time slot to suppress interference to co-existing systems and ensure the stability of all data queues, thereby adapting to non-uniform communication demands [28].

B. Link Model

In the n-th time slot, the binary variable $f_{s,k}[n]$ is introduced to indicate whether the beam of LEO satellite s illuminates the k-th cell. If $f_{s,k}[n] = 1$, the k-th cell is illuminated by LEO satellite s during the n-th time slot, otherwise, it is not illuminated. These variables form the coverage for cells $\mathbf{f}_s[n] \in \{0,1\}^{|\mathcal{K}|\times 1}$ by LEO satellite s during the

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TABLE I
LIST OF PRIMARY NOTATIONS

Notations	Definitions
W	The bandwidth of each frequency segment
$\mathcal{L}$	The set of the frequency segments
$\mathcal{S}$	The set of the LEO satellites
$\mathcal{K}$	The set of the cells
N_b	The number of beams in each LEO satellite
$\Omega(s)$	The set of the cells covered by satellite s
$\Phi(k)$	The set of the LEO satellites that can serve the k-th cell
$\mathcal{K}_{\mathrm{G}}$	The set of the cells where the GSO ground stations exist
$\mathcal{L}_k$	The set of the overlapping frequency segments between the LEO and GSO satellites for the ground station in the k-th cell
T_0	The duration of each time slot
$\mathcal{N}$	The set of the time slots
$f_{s,k}[n]$	The binary variable indicating whether the beam of LEO satellite s illuminates the k-th cell
$p_{s,k,l}[n]$	The power allocated by LEO satellite s in the l-th frequency segment for the beam illuminating the k-th cell
$h_{s,k,j}[n]$	The channel coefficient between the receiver in the k-th cell and the beam illuminating the j-th cell from LEO satellite s
$I_{s,k,l}[n]$	The interference encountered by the l-th frequency segment of the beam illuminating the k-th cell in LEO satellite s
$\gamma_{s,k,l}[n]$	The SINR for the l-th frequency segment of the beam illuminating the k-th cell in LEO satellite s
N_0	The noise spectral power density
$R_{s,k}[n]$	The achievable data rate for the beam in LEO satellite s illuminating the k-th cell
$g_{s,k,j}[n]$	The channel coefficient from the beam in LEO satellite s serving the j-th cell to the GSO ground station in the k-th cell
$Z_{k,l}[n]$	The interference in the l-th frequency segment encountered by the GSO station in the k-th cell
P_{I}	The power consumption of each laser ISL terminal
R_{I}	The achievable data rate of each laser ISL
M_0	The size of each data packet
$a_{s,k}[n]$	The number of arriving packets for the k-th cell on LEO satellite s
$q_{s,k}[n]$	The queue length for packets to the k-th cell on LEO satellite s at the beginning of the n-th time slot
$b_{r,s,k}[n]$	The number of data packets transferred from LEO satellite r to LEO satellite s destined for the k-th cell
$c_{r,s}[n]$	The number of bidirectionally transferred packets through the laser ISL between LEO satellites r and s for load balancing
$d_{s,k}[n]$	The overall change in the queue length $q_{s,k}[n]$ for load balancing
$x_{s,k}[n]$	The number of transmitted packets from LEO satellite s to the k-th cell
$E_s[n]$	The energy consumption of LEO satellite s
$E_{\mathrm{T},s}[n]$	The energy consumed for data transmission in LEO satellite s
$E_{\mathrm{B},s}[n]$	The energy expended for load balancing in LEO satellite s
$\mathcal{E}$	The long-term average energy consumption for all the LEO satellites
V	The trade-off coefficient for queue stability and energy consumption

n-th time slot, known as the BH pattern of LEO satellite s, and thereby the BH pattern matrix $\mathbf{F}[n] \in \{0,1\}^{|\mathcal{K}|\times|\mathcal{S}|}$ for all LEO satellites can be obtained. Similarly, the power allocation during the n-th time slot of LEO satellite s in the l-th frequency segment of the beam illuminating the k-th cell is denoted as $p_{s,k,l}[n]$, subsequently constituting the resource management tensor $\mathbf{P}[n] \in \mathbb{R}^{|\mathcal{L}|\times|\mathcal{K}|\times|\mathcal{S}|}$ for all satellites across cells and frequency segments in the n-th time slot.

In LEO satellite downlink communications, the transmitting antenna gain of the LEO satellite is highly directional to ensure the service quality to the targeted cell and reduce the interference to other cells [13]. Conversely, for the receiver in the cell, because it can receive signals from multiple LEO satellites simultaneously, the receiving antenna gain is identical for all LEO satellites that can serve this cell [20]. Moreover, the interference from satellites whose coverage does not contain this cell is negligible due to the low receiver elevation angle and long propagation distance. Specifically, the channel coefficient between the receiver in the k-th cell and the beam illuminating the j-th cell from LEO satellite s is denoted as $h_{s,k,j}[n]$. Although the Doppler shift can be compensated with the satellite orbital information, other factors such as free-space path loss, antenna gains, multipath propagation, obstructions, and atmospheric attenuation still influence the channel coefficients $h_{s,k,j}[n]$ [12], [15], [16], [37], [38], [39]. However, the proposed method does not rely on the specific form of the channel, as demonstrated in the subsequent analysis. Therefore, during the n-th time slot, the l-th frequency segment of the beam illuminating the k-th cell in LEO satellite s encounters the interference not only from other beams of the same satellite but also from other satellites in $\Phi(k)$, which is calculated as

$$I_{s,k,l}[n] = \sum_{j\in\mathcal{K}\setminus\{k\}} |h_{s,k,j}[n]|^2 p_{s,j,l}[n] f_{s,j}[n] + \sum_{r\in\Phi(k)\setminus\{s\}} \sum_{j\in\mathcal{K}} |h_{r,k,j}[n]|^2 p_{r,j,l}[n] f_{r,j}[n], \quad (1)$$

where the first and second terms represent the intra- and inter-satellite interference, respectively. Therefore, the signal-to-interference-plus-noise ratio (SINR) for the l-th frequency segment of the beam illuminating the k-th cell in LEO satellite s is expressed as

$$\gamma_{s,k,l}[n] = \frac{|h_{s,k,k}[n]|^2 p_{s,k,l}[n] f_{s,k}[n]}{N_0 W + I_{s,k,l}[n]}, \quad (2)$$

where N_0 is the noise spectral power density. Subsequently, the achievable data rate for the beam in LEO satellite s illuminating the k-th cell can be obtained by aggregating the data rate on each frequency segment as

$$R_{s,k}[n] = f_{s,k}[n] \sum_{l\in\mathcal{L}} W \log_2(1+\gamma_{s,k,l}[n]). \quad (3)$$

For the GSO ground station in the k-th cell, the channel coefficient from the beam in LEO satellite s serving the j-th cell is denoted as $g_{s,k,j}[n]$ during the n-th time slot. Moreover, $g_{s,k,j}[n]$ varies significantly for different satellites due to the highly concentrated gain of the receiving antenna at the GSO ground station. Therefore, the interference in the l-th frequency segment encountered by this GSO station from LEO satellite communications is given by

$$Z_{k,l}[n] = \sum_{s\in\mathcal{S}} \sum_{j\in\mathcal{K}} |g_{s,k,j}[n]|^2 p_{s,j,l}[n] f_{s,j}[n]. \quad (4)$$

Moreover, the channel coefficients $h_{s,k,j}[n]$ and $g_{s,k,j}[n]$ are assumed to remain constant during the n-th time slot. Considering the motion of LEO satellites, these coefficients are updated among various time slots.

Additionally, the laser ISL between LEO satellites is established between two space-borne laser terminals. Due to the abundant bandwidth and concentrated energy of laser ISLs, the data rate is primarily limited by the receiver sensitivity, tracking mechanism, and signal processing capabilities of the space-borne laser terminals, rather than by bandwidth or power [40]. In addition, the variation of inter-satellite distance is insignificant over a short time. Therefore, constant power and data rate are assumed to transfer data packets between satellites. Meanwhile, since the transmitter and receiver share the same optical path in space-borne laser terminals [40], the laser ISL results in symmetric power consumption at

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both ends, where the power consumption of one laser ISL terminal is denoted as P_{I}, and the achievable data rate of its bidirectional transmission is represented by R_{I}.

C. Queue Model

In each time slot, data packets with fixed volume M_0 arrive at LEO satellites for transmission to the cells. Specifically, the number of arriving packets destined for the k-th cell on LEO satellite s during the n-th time slot is denoted as $a_{s,k}[n]$, with an upper bound of $a_{\max}$. Subsequently, these packets queue on LEO satellite s for transmission to the targeted cell. At the beginning of the n-th time slot, the queue length for packets intended for the k-th cell on LEO satellite s is indicated by $q_{s,k}[n]$. Besides, the set of all queue lengths in the n-th time slot is denoted as $\mathbf{Q}[n]$.

Additionally, achieving load balancing among LEO satellites via laser ISLs is crucial due to the non-uniform geographical distribution of communication demands and the varying interference levels imposed by different LEO satellites on GSO ground stations within a given cell. The number of data packets transferred from LEO satellite r to LEO satellite s during the n-th time slot destined for the k-th cell is denoted as $b_{r,s,k}[n]$. If $b_{r,s,k}[n] > 0$, it indicates that packets are transferred from LEO satellite r to LEO satellite s, otherwise, the transfer is reversed. Then, the load balancing tensor $\mathbf{B}[n] \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{S}|\times|\mathcal{K}|}$ among all LEO satellites is constructed.

Therefore, with the determined load balancing scheme $\mathbf{B}[n]$, the amount of bidirectionally transferred packets through the laser ISL between LEO satellite r and LEO satellite s for load balancing during the n-th time slot is given by

$$c_{r,s}[n] = \sum_{k\in\mathcal{K}} |b_{r,s,k}[n]|. \tag{5}$$

Besides, since the laser ISLs also undertake other tasks, the number of packets transferred for load balancing on each laser ISL is limited by an upper bound of $c_{\max}$. Moreover, the overall change in queue length $q_{s,k}[n]$ for load balancing can be determined with $\mathbf{B}[n]$ by

$$d_{s,k}[n] = \sum_{r\in\Phi(k)} b_{r,s,k}[n], \tag{6}$$

where $d_{s,k}[n] > 0$ represents the queue length increases after load balancing, whereas $d_{s,k}[n] < 0$ denotes a decrease.

For the downlink transmission of LEO satellite, the number of transmitted packets is constrained by both the achievable data rate and the current queue length. Therefore, the number of transmitted packets from LEO satellite s to the k-th cell during the n-th time slot can be expressed as

$$x_{s,k}[n] = \min\left\{R_{s,k}[n]T_0/M_0,\ q_{s,k}[n] + d_{s,k}[n]\right\}, \tag{7}$$

which depends on the smaller one between the number of data packets that can be transmitted with the achievable data rate $R_{s,k}[n]$ and the number of data packets waiting in the queue after load balancing. Therefore, proper resource management $\mathbf{P}[n]$ and load balancing $\mathbf{B}[n]$ are crucial for the satellite network. Additionally, since the decisions for the BH pattern, resource management, and load balancing are determined at the beginning of the n-th time slot, the stochastic number of arriving packets $a_{s,k}[n]$ is uncertain at this moment and therefore not included in (7), whereas the impact of load balancing is considered.

Consequently, the update of the queue length at the end of the n-th time slot is given by

$$q_{s,k}[n+1] = q_{s,k}[n] + d_{s,k}[n] + a_{s,k}[n] - x_{s,k}[n], \tag{8}$$

which includes the arriving packets, transferred packets in load balancing, and transmitted packets for satellite-terrestrial communications.

III. Problem Formulation and Transformation

In Section III-A, the optimization objective of long-term average energy consumption is defined, while the overall optimization problem with BH pattern, resource management, and load balancing is formulated. Subsequently, in Section III-B, the long-term problem is transformed into optimization within each time slot by the Lyapunov drift-plus-penalty method.

A. Optimization Problem Formulation

Energy consumption is a crucial metric in LEO satellite communications, where the energy consumption $E_s[n]$ of LEO satellite s comprises two components: the energy consumed by direct transmission to cells $E_{\mathrm{T},s}[n]$, and the energy expended via laser ISLs during load balancing $E_{\mathrm{B},s}[n]$. Therefore, $E_s[n] = E_{\mathrm{T},s}[n] + E_{\mathrm{B},s}[n]$, and the former can be directly obtained as

$$E_{\mathrm{T},s}[n] = \sum_{k\in\mathcal{K}}\sum_{l\in\mathcal{L}} p_{s,k,l}[n]T_0. \tag{9}$$

Moreover, the transfer time required for load balancing through the laser ISL between LEO satellite r and LEO satellite s can be calculated as

$$t_{r,s}[n] = \frac{M_0 c_{r,s}[n]}{R_{\mathrm{I}}}. \tag{10}$$

Therefore, the energy consumption for load balancing in LEO satellite s via laser ISLs is given by

$$E_{\mathrm{B},s}[n] = \sum_{r\in\mathcal{S}} P_{\mathrm{I}} t_{r,s}[n]. \tag{11}$$

Notably, the space-borne terminals are present at both ends of the laser ISL, the energy consumption of laser ISL should be calculated twice in satellites at both ends.

Due to the swift motion of LEO satellites and the time-varying communication demands, the expectation operator $\mathbb{E}\{\cdot\}$ is introduced, while the long-term average energy consumption is the main focus as

$$\mathcal{E} = \lim_{|\mathcal{N}|\to\infty} \frac{1}{|\mathcal{N}|}\mathbb{E}\left\{\sum_{n\in\mathcal{N}}\sum_{s\in\mathcal{S}} E_s[n]\right\}. \tag{12}$$

The optimization objective of this research is to minimize the long-term average energy consumption $\mathcal{E}$ of the LEO

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satellite network by optimizing the BH pattern $\mathbf{F}[n]$, resource management $\mathbf{P}[n]$, and load balancing $\mathbf{B}[n]$ in each time slot. Moreover, it should be ensured that the service capacity of the LEO satellite network reliably meets the communication demands of terrestrial cells. Meanwhile, the interference to co-existing GSO ground stations is also suppressed. Therefore, the overall optimization problem is formulated as

$$\text{(P0)}: \min_{\mathbf{F}[n],\mathbf{P}[n],\mathbf{B}[n],n\in\mathcal{N}} \mathcal{E} \tag{13}$$

$$\text{s.t.} \quad \lim_{|\mathcal{N}|\to\infty} \frac{1}{|\mathcal{N}|}\mathbb{E}\left\{|q_{s,k}[n]|\right\}=0,\ \forall\ s\in\mathcal{S},\ \forall\ k\in\mathcal{K}, \tag{13a}$$

$$f_{s,k}[n](1-f_{s,k}[n])=0,\quad \forall\ s\in\mathcal{S},\ \forall k\in\mathcal{K}, \tag{13b}$$

$$f_{s,k}[n]=0,\quad \forall\ s\in\mathcal{S},\ \forall\ k\notin\Omega(s), \tag{13c}$$

$$\sum_{k\in\mathcal{K}} f_{s,k}[n]\le N_b,\quad \forall\ s\in\mathcal{S}, \tag{13d}$$

$$p_{s,k,l}[n]\ge 0,\quad \forall\ s\in\mathcal{S},\ \forall\ k\in\mathcal{K},\ \forall\ l\in\mathcal{L}, \tag{13e}$$

$$\sum_{k\in\mathcal{K}}\sum_{l\in\mathcal{L}} p_{s,k,l}[n]\le P_{\max},\quad \forall\ s\in\mathcal{S}, \tag{13f}$$

$$\sum_{l\in\mathcal{L}_k} Z_{k,l}[n]\le Z_{\max},\quad \forall\ k\in\mathcal{K}_{\mathrm{G}}, \tag{13g}$$

$$b_{r,s,k}[n]+b_{s,r,k}[n]=0,\ \forall\ r,s\in\mathcal{S},\ \forall\ k\in\mathcal{K}, \tag{13h}$$

$$b_{r,s,k}[n]=0,\quad \forall\ r\in\mathcal{S},\ \forall\ s\notin\Phi(k), \tag{13i}$$

$$c_{r,s}[n]\le c_{\max},\quad \forall\ r,s\in\mathcal{S}, \tag{13j}$$

$$x_{s,k}[n]M_0\le R_{s,k}[n]T_0,\quad \forall\ s\in\mathcal{S},\ \forall\ k\in\mathcal{K}, \tag{13k}$$

$$x_{s,k}[n]\le q_{s,k}[n]+d_{s,k}[n],\ \forall\ s\in\mathcal{S},\ \forall\ k\in\mathcal{K}, \tag{13l}$$

where the above constraints can be categorized as

- **Queue stability constraint**: Constraint (13a) indicates that all queues should meet the mean rate stability condition, whose satisfaction ensures that the average arriving rate of communication demand does not exceed the average service rate as pointed in [28], thereby guaranteeing that the communication demands for each cell can be reliably satisfied by the LEO satellite network.
- **BH pattern constraints**: The BH pattern indicator variables are binary as specified by constraint (13b). Moreover, constraints (13c) and (13d) stipulate that each LEO satellite can only serve cells within its coverage area and the number of served cells is limited by the number of equipped beams, respectively.
- **Resource management constraints**: Constraint (13e) requires the transmission power to be non-negative. Besides, constraint (13f) sets the upper bound of the downlink transmission power for each satellite to $P_{\max}$, which is sufficient for ensuring the queue stability.
- **Interference suppression constraint**: To suppress the interference to the co-existing system, the sum of interference to the GSO ground station on all frequency segments is suppressed not to exceed $Z_{\max}$, as indicated in constraint (13g).
- **Load balancing constraints**: Constraint (13h) mandates that the amount of incoming and outgoing packets for satellites at both ends of the laser ISL should be equal. Moreover, constraint (13i) dictates that packets destined for the k-th cell can only be transferred among satellites within $\Phi(k)$. Besides, the number of transferred packets per laser ISL during load balancing is limited by constraint (13j).
- **Transmission constraints**: Constraints (13k) and (13l) ensure the compliance with the condition in (7) regarding the number of transmitted packets.

B. Problem Transformation by Lyapunov Optimization

The problem (P0) aims to minimize the long-term average energy consumption while ensuring the mean rate stability of the queues on LEO satellites, i.e., preventing the continuous growth of queue length. However, solving this problem directly is challenging. Specifically, the objective function and constraint (13a) indicate a focus on long-term effects, while solving from a long-term perspective poses computational complexity challenges and requires substantial information. Therefore, a dynamic algorithm is necessary to balance the queue length and energy consumption, based on the queue length and channel coefficients in the current time slot. In particular, the Lyapunov function is introduced to characterize the queue length as

$$L_{\mathrm{q}}[n]=\frac{1}{2}\sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}} q_{s,k}^2[n]. \tag{14}$$

Moreover, to reflect the impact of the optimization on the queue length in each time slot with the given queue lengths $\mathbf{Q}[n]$, the Lyapunov drift is defined as

$$\Delta L_{\mathrm{q}}[n]=\mathbb{E}\{L_{\mathrm{q}}[n+1]-L_{\mathrm{q}}[n]|\mathbf{Q}[n]\}. \tag{15}$$

Additionally, for the queue stability, the Lyapunov drift should be as small as possible in each time slot. Besides, for the objective function in problem (P0), the energy consumption in each time slot is incorporated as a penalty term into the Lyapunov drift, thereby forming the transformed Lyapunov optimization problem for each time slot as

$$\text{(P0-1)}: \min_{\mathbf{F}[n],\mathbf{P}[n],\mathbf{B}[n]} \frac{\Delta L_{\mathrm{q}}[n]}{L_0}+\frac{V}{E_0}\mathbb{E}\left\{\sum_{s\in\mathcal{S}} E_s[n]\Big|\mathbf{Q}[n]\right\} \tag{16}$$
$$\text{s.t.} \quad (13b)-(13l),$$

where V is an adjustable coefficient to balance the queue length and energy consumption. To aggregate the terms in different units, the normalization factors L_0 and E_0 are applied.

However, solving problem (P0-1) remains a challenge. Therefore, minimizing the upper bound of the objective function is considered as an alternative solution [28]. Firstly, the boundedness of $a_{s,k}[n]$, $x_{s,k}[n]$, and $d_{s,k}[n]$ in each time slot is established as

$$0\le a_{s,k}[n]\le a_{\max}, \tag{17a}$$

$$0\le x_{s,k}[n]\le x_{\max}, \tag{17b}$$

$$-d_{\max}\le d_{s,k}[n]\le d_{\max}, \tag{17c}$$

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where (17a) holds due to the maximum number of arriving packets. Moreover, because of the constraints (13f) and (13j), the number of transmitted packets $x_{s,k}[n]$ and the absolute value of the change in each queue $|d_{s,k}[n]|$ are also bounded, whose upper bounds are denoted as $x_{\max}$ and $d_{\max}$, respectively. Based on the boundedness of these variables, the upper bound of the Lyapunov drift can be derived as

$$\begin{aligned}\Delta L_{\mathrm{q}}[n] &= \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}\mathbb{E}\Big\{\frac{1}{2}a_{s,k}^2[n]+\frac{1}{2}x_{s,k}^2[n]+\frac{1}{2}d_{s,k}^2[n] \\ &\quad + a_{s,k}[n](d_{s,k}[n]-x_{s,k}[n])-d_{s,k}[n]x_{s,k}[n] \\ &\quad + q_{s,k}[n](a_{s,k}[n]+d_{s,k}[n]-x_{s,k}[n])\big|\mathbf{Q}[n]\Big\} \quad (18) \\ &\le \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}\mathbb{E}\Big\{\frac{1}{2}a_{\max}^2+\frac{1}{2}x_{\max}^2+\frac{1}{2}d_{s,k}^2[n] \\ &\quad + a_{\max}d_{\max}+d_{\max}x_{\max}+q_{s,k}[n]a_{s,k}[n] \\ &\quad + q_{s,k}[n](d_{s,k}[n]-x_{s,k}[n])\big|\mathbf{Q}[n]\Big\} \quad (18a) \\ &= \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}\Big(\frac{1}{2}a_{\max}^2+\frac{1}{2}x_{\max}^2 \\ &\quad + (a_{\max}+x_{\max})d_{\max}+q_{s,k}[n]\mathbb{E}\{a_{s,k}[n]\}+\mathbb{E}\Big\{ \\ &\quad q_{s,k}[n](d_{s,k}[n]-x_{s,k}[n])+\frac{1}{2}d_{s,k}^2[n]\big|\mathbf{Q}[n]\Big\}\Big) \quad (18b) \\ &= \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}\Big(\mathbb{E}\Big\{q_{s,k}[n](d_{s,k}[n]-x_{s,k}[n]) \\ &\quad + \frac{1}{2}d_{s,k}^2[n]\big|\mathbf{Q}[n]\Big\}+q_{s,k}[n]\mathbb{E}\{a_{s,k}[n]\}\Big)+\Upsilon_1, \quad (18c)\end{aligned}$$

where (18a) is derived by scaling the cross terms that are independent of $\mathbf{Q}[n]$ and the quadratic terms to their upper bounds, while $d_{s,k}^2[n]$ is retained as a regularization term. Moreover, because the distribution of $a_{s,k}[n]$ is independent of other variables, its expectation is separated in (18b). Besides, the upper bound (18c) is obtained after rearranging all terms, while Υ_1 represents the constant term.

Therefore, by eliminating the constant term Υ_1 and the term $q_{s,k}[n]\mathbb{E}\{a_{s,k}[n]\}$ unrelated to the optimization variables from (18c), while discarding the conditional expectation based on the concept of opportunistic optimization, a modified objective with the given coefficient V can be derived as

$$\begin{aligned}\Lambda(\mathbf{F}[n],\mathbf{P}[n],\mathbf{B}[n];V) &= \sum_{s\in\mathcal{S}}\Big(\frac{1}{L_0}\sum_{k\in\mathcal{K}}\big(q_{s,k}[n](d_{s,k}[n] \\ &\quad -x_{s,k}[n])+\frac{1}{2}d_{s,k}^2[n]\big)+\frac{V}{E_0}E_s[n]\Big), \quad (19)\end{aligned}$$

where the first term assigns greater weight to queues with longer lengths $q_{s,k}[n]$ in the objective, making them more likely to decrease $d_{s,k}[n]$ by transferring data packets to other satellites through load balancing, and to increase the number of transmitted data packets $x_{s,k}[n]$, while $d_{s,k}^2[n]/2$ serves as a regularization term. Meanwhile, the second term prioritizes minimizing the energy consumption $E_s[n]$. Consequently, the problem (P0-1) is further transformed into the following optimization problem as

$$\begin{aligned}(\text{P0-2}): &\min_{\mathbf{F}[n],\mathbf{P}[n],\mathbf{B}[n]} \Lambda(\mathbf{F}[n],\mathbf{P}[n],\mathbf{B}[n];V) \quad (20)\\ &\text{s.t.} \quad (13b)-(13l).\end{aligned}$$

In addition, it is assumed that the channel coefficients $h_{s,k,j}[n]$ and $g_{s,k,j}[n]$, along with the number of arriving packets $a_{s,k}[n]$, follow a stationary distribution. Since the maximum transmission power in (13f) is sufficient, an optimal policy can be designed solely based on the current channel coefficients to achieve the minimum energy consumption $\mathcal{E}_{\mathrm{opt}}$ while satisfying the constraints in (P0), as analyzed in [28]. Meanwhile, the performance gap between the results obtained by the Lyapunov drift-plus-penalty method and the possible optimal solution is ensured by the following proposition.

Proposition 1: If the channel coefficients can be modeled as Markov processes, given the coefficient $V > 0$, the performance gap between the $\mathcal{E}$ obtained by solving problem (P0-2) and the optimal $\mathcal{E}_{\mathrm{opt}}$ is given by

$$\mathcal{E} \le \mathcal{E}_{\mathrm{opt}} + \mathcal{O}(1/V). \quad (21)$$

If the Slater condition is satisfied, meaning that the average service rate of the satellite network is strictly greater than the average packet arrival rate while still meeting all constraints, then the average queue length is bounded as

$$\lim_{|\mathcal{N}|\to\infty}\frac{1}{|\mathcal{N}|}\mathbb{E}\left\{\sum_{n\in\mathcal{N}}\sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}q_{s,k}[n]\right\} \le \mathcal{O}(V). \quad (22)$$

Proof: See Appendix A.

According to Little's Law, the average queue length is proportional to the average queueing latency in steady state. Therefore, *Proposition* 1 characterizes the fundamental trade-off between energy consumption and queueing latency. As the control parameter V increases, the energy consumption of the satellite network approaches its optimal value; however, this comes at the cost of a larger average queue length, which corresponds to a longer queueing latency.

Although the original long-term optimization problem is transformed into optimization in each time slot with guaranteed performance, solving the problem (P0-2) remains challenging. Specifically, the BH pattern, resource management, and load balancing are mutually coupled, while the joint optimization of $\mathbf{F}[n]$, $\mathbf{P}[n]$, and $\mathbf{B}[n]$ introduces significant computational complexity. To address this issue, the BCD algorithm is adopted, where each set of variables is optimized sequentially while keeping the others fixed.

IV. Problem Solution for Energy Minimization

The optimization of problem (P0-2) is discussed in this section, and since the scope of discussion is confined to each time slot, the index $[n]$ of variables is omitted for simplicity. The BH pattern $\mathbf{F}$, resource management $\mathbf{P}$, and load balancing $\mathbf{B}$ are optimized alternately with BCD algorithm in **Algorithm** 1. Specifically, the BH pattern is determined by the MPMM approach in Section IV-A, which integrates the MP method for

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Algorithm 1 BCD Algorithm for $\mathbf{F}, \mathbf{P}$, and $\mathbf{B}$

Require: Channel coefficients $h_{s,k,j}$ and $g_{s,k,j}$, queue length $\mathbf{Q}$.
Ensure: Optimized BH pattern $\mathbf{F}^{\text{opt}}$, resource management $\mathbf{P}^{\text{opt}}$, and load balancing $\mathbf{B}^{\text{opt}}$.
Init: Initial value $\mathbf{F}^{\{0\}}$, $\mathbf{P}^{\{0\}}$, $\mathbf{B}^{\{0\}}$, $i \leftarrow 0$
repeat
  Given fixed $\mathbf{P}^{\{i\}}$ and $\mathbf{B}^{\{i\}}$, minimize (19) by optimizing $\mathbf{F}$ with the MPMM approach, and acquire $\mathbf{F}^{\{i+1\}}$.
  Given fixed $\mathbf{F}^{\{i+1\}}$ and $\mathbf{B}^{\{i\}}$, minimize (19) by optimizing $\mathbf{P}$ with the SCA method, and obtain $\mathbf{P}^{\{i+1\}}$.
  Given fixed $\mathbf{F}^{\{i+1\}}$ and $\mathbf{P}^{\{i+1\}}$, minimize (19) with $\mathbf{B}$ by quadratic programming, thereby achieving $\mathbf{B}^{\{i+1\}}$.
  $i \leftarrow i+1$
until $\Lambda(\mathbf{F}^{\{i\}}, \mathbf{P}^{\{i\}}, \mathbf{B}^{\{i\}}; V)$ is convergence
$\mathbf{F}^{\text{opt}} \leftarrow \mathbf{F}^{\{i\}}$, $\mathbf{P}^{\text{opt}} \leftarrow \mathbf{P}^{\{i\}}$, $\mathbf{B}^{\text{opt}} \leftarrow \mathbf{B}^{\{i\}}$.

the binary constraint and the MM algorithm for complicated objectives and constraints. Moreover, the resource management for joint optimization of power and frequency segment is tackled with SCA in Section IV-B, while load balancing is achieved by quadratic programming in Section IV-C. Finally, the computational complexity is analyzed in Section IV-D.

A. MPMM for BH Pattern

Given fixed resource management $\mathbf{P}^*$ and load balancing $\mathbf{B}^*$, the optimization for the BH pattern $\mathbf{F}$ is formulated as

$$\text{(P1)}: \min_{\mathbf{F}} \ \Lambda(\mathbf{F}, \mathbf{P}^*, \mathbf{B}^*; V) \tag{23}$$
$$\text{s.t.} \quad \text{(13b), (13c), (13d), (13g), (13k), and (13l).}$$

Due to the presence of constraint (13b), problem (P1) constitutes a mixed-integer nonlinear programming problem, for which obtaining the global optimal solution in polynomial time is difficult. To achieve an efficient solution, the constraint (13b) is incorporated into the objective function as an MP term to obtain the approximate solution by a series of rounds [41], and the introduced MP term in the θ-th round is expressed as

$$J^{(\theta)}(\mathbf{F}) = \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}} \alpha_{s,k}^{(\theta)} f_{s,k}(1-f_{s,k}) + \beta^{(\theta)} \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}} f_{s,k}^2(1-f_{s,k})^2, \tag{24}$$

where the first term corresponds to the Lagrange multiplier, and the second term represents the penalty function. These terms are combined to enhance the numerical stability of the algorithm, while the positive coefficients $\alpha_{s,k}^{(\theta)}$ and $\beta_{s,k}^{(\theta)}$ update in each round. Moreover, binary constraint (13b) is further relaxed as

$$0 \le f_{s,k} \le 1, \quad \forall s \in \mathcal{S}, \quad \forall k \in \mathcal{K}. \tag{25}$$

Consequently, in the θ-th round, the problem is formulated as

$$\text{(P1-}\theta\text{)}: \min_{\mathbf{F}[n]} \ \Lambda(\mathbf{F}, \mathbf{P}^*, \mathbf{B}^*; V) + J^{(\theta)}(\mathbf{F}) \tag{26}$$
$$\text{s.t.} \quad \text{(13c), (13d), (13g), (13k), (13l), and (25).}$$

Additionally, the update rules in each round for $\alpha_{s,k}^{(\theta)}$ and $\beta_{s,k}^{(\theta)}$ are given by

$$\alpha_{s,k}^{(\theta+1)} = \alpha_{s,k}^{(\theta)} + 2\beta^{(\theta)} f_{s,k}^{(\theta)}(1-f_{s,k}^{(\theta)}), \tag{27a}$$
$$\beta^{(\theta+1)} = \rho\beta^{(\theta)}, \tag{27b}$$

where $f_{s,k}^{(\theta)}$ is the solution of problem (P1-θ) in the θ-th round while forming $\mathbf{F}^{(\theta)}$, and $\rho \ge 1$ is adopted to gradually strengthen the penalty coefficient so that the algorithm converges to a feasible solution.

However, solving (P1-θ) remains complicated since the MP term $J^{(\theta)}(\mathbf{F})$ is high-order and non-convex. Moreover, for the achievable data rate $R_{s,k}$ in constraint (13k), the second partial derivative with respect to $f_{s,k}$ can be calculated as

$$\frac{\partial^2 R_{s,k}}{\partial f_{s,k}^2} = \sum_{l\in\mathcal{L}} \frac{W|h_{s,k,k}|^2 p_{s,k,l}(2+\gamma_{s,k,l})}{\ln 2(N_0W + I_{s,k,l})(1+\gamma_{s,k,l})^2} \ge 0, \tag{28}$$

which reveals that $R_{s,k}$ is non-concave with respect to $\mathbf{F}$. Consequently, constraint (13k) is also non-convex.

To address this issue, the MM algorithm is introduced to iteratively solve problem (P1-θ). In each iteration, given the current initial point $\widehat{\mathbf{F}}$, a surrogate function tangent to the original function at this point is selected, allowing the problem to be solved in a more concise form [42]. It can be verified that the result obtained through this approach eventually converges to the Karush-Kuhn-Tucker (KKT) solution of problem (P1-θ). Specifically, given the initial point $\widehat{\mathbf{F}}$, a surrogate function for the MP term $J^{(\theta)}(\mathbf{F})$ can be constructed as

$$\overline{J}^{(\theta)}(\mathbf{F};\widehat{\mathbf{F}}) = \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}} \alpha_{s,k}^{(\theta)} \left((1-2\widehat{f}_{s,k})f_{s,k} + \widehat{f}_{s,k}^2\right) + \beta^{(\theta)} \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}} \left((1-2\widehat{f}_{s,k})f_{s,k} + \widehat{f}_{s,k}^2\right)^2, \tag{29}$$

which is a convex upper bound of $J^{(\theta)}(\mathbf{F})$ while remaining tangent to $J^{(\theta)}(\mathbf{F})$ at the point $\widehat{\mathbf{F}}$. For the surrogate function of the achievable data rate $R_{s,k}$, the following auxiliary variables dependent on $\mathbf{F}$ are introduced for simplicity as

$$\phi_{s,k,l}^{\text{F}}(\mathbf{F}) = \frac{\gamma_{s,k,l}(\mathbf{F})}{1+\gamma_{s,k,l}(\mathbf{F})}, \tag{30a}$$
$$\zeta_{s,k,l}^{\text{F}}(\mathbf{F}) = \log_2\left(1+\gamma_{s,k,l}(\mathbf{F})\right) - \phi_{s,k,l}^{\text{F}}(\mathbf{F})\log_2\left(\gamma_{s,k,l}(\mathbf{F})\right), \tag{30b}$$

where $\gamma_{s,k,l}(\mathbf{F})$ indicates that $\mathbf{F}$ is the only independent variable with other variables fixed. Subsequently, the corresponding surrogate function is then obtained as

$$\underline{R}_{s,k}^{\text{F}}(\mathbf{F};\widehat{\mathbf{F}}) = \sum_{l\in\mathcal{L}} W\Bigg(f_{s,k}\Big(\phi_{s,k,l}^{\text{F}}(\widehat{\mathbf{F}})\log_2\left(e\gamma_{s,k,l}(\widehat{\mathbf{F}})\right) + \zeta_{s,k,l}^{\text{F}}(\widehat{\mathbf{F}})\Big) - \log_2(e)\phi_{s,k,l}^{\text{F}}(\widehat{\mathbf{F}})\gamma_{s,k,l}(\widehat{\mathbf{F}})\frac{N_0W + I_{s,k,l}(\mathbf{F})}{|h_{s,k,k}|^2 p_{s,k,l}^*}\Bigg) - \kappa(f_{s,k}-\widehat{f}_{s,k})^2, \tag{31}$$

where $\mathbf{F}$ is the only independent variables in $I_{s,k,l}(\mathbf{F})$, similar to $\gamma_{s,k,l}(\mathbf{F})$, and $\kappa > 0$ is the regularization coefficient. Moreover, it can be certified that $\underline{R}_{s,k}^{\text{F}}(\mathbf{F};\widehat{\mathbf{F}})$ is concave in $\mathbf{F}$, and the following proposition holds.

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Proposition 2: The surrogate function $\underline{R}^{\mathrm{F}}_{s,k}(\mathbf{F};\widehat{\mathbf{F}})$ is a lower bound of $R_{s,k}(\mathbf{F})$, and is tangent to $R_{s,k}(\mathbf{F})$ at the point $\widehat{\mathbf{F}}$.

Proof: See Appendix B.

Consequently, during the θ-th round in problem (P1-θ), if the solution obtained by MM algorithm at the ψ-th iteration is denoted as $\tilde{\mathbf{F}}^{[\psi]}$, then the problem to be solved at the $(\psi+1)$-th iteration is further formulated as

$$\begin{aligned}(\text{P1-}\theta\text{-}\psi):\ &\min_{\mathbf{F}[n]}\ \Lambda(\mathbf{F},\mathbf{P}^*,\mathbf{B}^*;V)+\overline{J}^{(\theta)}(\mathbf{F};\tilde{\mathbf{F}}^{[\psi]}) \qquad (32)\\ &\text{s.t.}\ \ (13c),(13d),(13g),(13l),\ \text{and}\ (25),\\ &\quad x_{s,k}M_0\le \underline{R}^{\mathrm{F}}_{s,k}(\mathbf{F};\tilde{\mathbf{F}}^{[\psi]})T_0,\ \ \forall\ s\in\mathcal{S},\\ &\qquad\qquad \forall\ k\in\mathcal{K}, \qquad (32a)\end{aligned}$$

where (32a) is derived by replacing $R_{s,k}$ in (13k) with its surrogate function $\underline{R}^{\mathrm{F}}_{s,k}(\mathbf{F};\tilde{\mathbf{F}}^{[\psi]})$. As a result, the problem (P1-θ-ψ) becomes a convex problem that can be directly solved with optimization toolbox such as GUROBI.

Algorithm 2 MPMM for BH Pattern $\mathbf{F}$

Require: $\mathbf{F}^{\{i\}}$, $\mathbf{P}^{\{i\}}$, $\mathbf{B}^{\{i\}}$, round and iteration numbers Θ, Ψ.
Ensure: $\mathbf{F}^{\{i+1\}}$.
1: **Init:** initial point $\mathbf{F}^{(0)}$, coefficients $\alpha^{(1)}_{s,k}$, $\beta^{(1)}$, and ρ.
2: $\mathbf{P}^*\leftarrow\mathbf{P}^{\{i\}}$, $\mathbf{B}^*\leftarrow\mathbf{B}^{\{i\}}$.
3: **for** round $\theta=1$ to Θ **do**
4: $\quad\tilde{\mathbf{F}}^{[0]}\leftarrow\mathbf{F}^{(\theta-1)}$.
5: $\quad$**for** iteration $\psi=1$ to Ψ **do**
6: $\qquad$Given $\tilde{\mathbf{F}}^{[\psi-1]}$, solve (P1-$\theta$-$\psi$) and obtain $\tilde{\mathbf{F}}^{[\psi]}$.
7: $\qquad$Update (P1-θ-ψ) based on $\tilde{\mathbf{F}}^{[\psi]}$.
8: $\quad$**end for**
9: $\quad\mathbf{F}^{(\theta)}\leftarrow\tilde{\mathbf{F}}^{[\Psi]}$, update $\alpha^{(\theta+1)}_{s,k}$, $\beta^{(\theta+1)}$ in (P1-θ).
10: **end for**
11: Quantize $\mathbf{F}^{(\Theta)}$ into a binary matrix.
12: **if** $\Lambda(\mathbf{F}^{(\Theta)},\mathbf{P}^*,\mathbf{B}^*;V)\le\Lambda(\mathbf{F}^{\{i\}},\mathbf{P}^*,\mathbf{B}^*;V)$ **then**
13: $\quad\mathbf{F}^{\{i+1\}}\leftarrow\mathbf{F}^{(\Theta)}$.
14: **else**
15: $\quad\mathbf{F}^{\{i+1\}}\leftarrow\mathbf{F}^{\{i\}}$.
16: **end if**

The details for optimizing BH pattern $\mathbf{F}$ are provided in **Algorithm 2**. The MP method is employed to optimize the binary variables through multiple rounds, as outlined in Steps 1∼10. Meanwhile, within each round, the MM algorithm is further utilized to iteratively solve the original problem. Once the number of iterations reaches the upper limit Ψ, $\tilde{\mathbf{F}}^{[\Psi]}$ is taken as the solution for the θ-th round, denoted by $\mathbf{F}^{(\theta)}$. Subsequently, $J^{(\theta+1)}(\mathbf{F})$ is updated accordingly, and the problem (P1-θ) is subsequently adjusted to initiate the $(\theta+1)$-th round for solving the problem (P1) until reach the round limit Θ. Finally, the result $\mathbf{F}^{(\Theta)}$ is quantized, and the superior one between $\mathbf{F}^{(\Theta)}$ and $\mathbf{F}^{\{i\}}$ is selected, ensuring a continuous reduction in the value of $\Lambda(\mathbf{F},\mathbf{P},\mathbf{B};V)$.

B. SCA for Resource Management

With the fixed BH pattern $\mathbf{F}^*$ and load balancing $\mathbf{B}^*$, the subproblem on resource management for joint optimization of power and frequency segment is established as

$$\begin{aligned}(\text{P2}):\ &\min_{\mathbf{P}}\ \Lambda(\mathbf{F}^*,\mathbf{P},\mathbf{B}^*;V) \qquad (33)\\ &\text{s.t.}\ \ (13e),(13f),(13g),(13k),\ \text{and}\ (13l),\end{aligned}$$

where the objective function is linear with $\mathbf{P}$. For the achievable rate $R_{s,k}$, the following expression can be derived as

$$\frac{\partial^2 R_{s,k}}{\partial p^2_{s,j,l}}=\frac{W f_{s,k}|h_{s,k,j}|^4 f^2_{s,j}\gamma_{s,k,l}(2+\gamma_{s,k,l})}{\ln 2(I_{s,k,l}+N_0W)^2(1+\gamma_{s,k,l})^2}\ge 0, \qquad (34)$$

which indicates that the constraint (13k) is non-convex in subproblem (P2). Therefore, the SCA technique is employed to transform the original non-convex problem into a series of convex problems by constructing an approximation function of $R_{s,k}(\mathbf{P})$ and updating it iteratively. Similarly, following auxiliary variables dependent on $\mathbf{P}$ are defined as

$$\eta^{\mathrm{P}}_{s,k,l}(\mathbf{P})=\frac{|h_{s,k,k}|^2 f^*_{s,k}}{N_0W+I_{s,k,l}(\mathbf{P})}, \qquad (35a)$$

$$\phi^{\mathrm{P}}_{s,k,l}(\mathbf{P})=\frac{\gamma_{s,k,l}(\mathbf{P})}{1+\gamma_{s,k,l}(\mathbf{P})}, \qquad (35b)$$

$$\begin{aligned}\zeta^{\mathrm{P}}_{s,k,l}(\mathbf{P})=&\log_2\left(1+\gamma_{s,k,l}(\mathbf{P})\right)\\ &-\phi^{\mathrm{P}}_{s,k,l}(\mathbf{P})\log_2\left(\gamma_{s,k,l}(\mathbf{P})\right). \qquad (35c)\end{aligned}$$

Hence, the approximation function can be expressed as

$$\begin{aligned}&\underline{R}^{\mathrm{P}}_{s,k}(\mathbf{P};\widehat{\mathbf{P}})\\ &=\sum_{l\in\mathcal{L}}W\Big(f^*_{s,k}\big(\phi^{\mathrm{P}}_{s,k,l}(\widehat{\mathbf{P}})\log_2\left(e\eta^{\mathrm{P}}_{s,k,l}(\widehat{\mathbf{P}})p_{s,k,l}\right)\\ &\quad+\zeta^{\mathrm{P}}_{s,k,l}(\widehat{\mathbf{P}})\big)-\log_2(e)\phi^{\mathrm{P}}_{s,k,l}(\widehat{\mathbf{P}})\eta^{\mathrm{P}}_{s,k,l}(\widehat{\mathbf{P}})\frac{N_0W+I_{s,k,l}(\mathbf{P})}{|h_{s,k,k}|^2}\Big), \qquad (36)\end{aligned}$$

where the first term is a logarithmic function of $p_{s,k,l}$, and the latter one is linear with $\mathbf{P}$, making $\underline{R}^{\mathrm{P}}_{s,k}(\mathbf{P};\widehat{\mathbf{P}})$ a concave function. Moreover, similar to the properties in *Proposition 2*, this approximation function serves as a lower bound for $R_{s,k}(\mathbf{P})$ and is tangent to it at the point $\widehat{\mathbf{P}}$.

As a result, assuming that the solution obtained for the ξ-th iteration of problem (P2) is $\mathbf{P}^{[\xi]}$, the subproblem for the $(\xi+1)$-th iteration can be expressed as

$$\begin{aligned}(\text{P2-}\xi):\ &\min_{\mathbf{P}}\ \Lambda(\mathbf{F}^*,\mathbf{P},\mathbf{B}^*;V) \qquad (37)\\ &\text{s.t.}\ \ (13e),(13f),(13g),\ \text{and}\ (13l),\\ &\quad x_{s,k}M_0\le\underline{R}^{\mathrm{P}}_{s,k}(\mathbf{P};\mathbf{P}^{[\xi]})T_0,\ \forall\ s\in\mathcal{S},\ \forall k\in\mathcal{K}, \qquad (37a)\end{aligned}$$

where the constraint (37a) is convex by substituting $R_{s,k}(\mathbf{P})$ with the approximation function. Therefore, the problem (P2-ξ) is a convex problem and can be solved conveniently. Consequently, the optimization of the resource management $\mathbf{P}$ with the SCA is presented in **Algorithm 3**, where the initial value $\mathbf{P}^{[0]}$ is set to the resource management matrix $\mathbf{P}^{\{i\}}$ obtained in the previous loop of the BCD algorithm.

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Algorithm 3 SCA for Resource Management $\mathbf{P}$

Require: $\mathbf{F}^{\{i+1\}}$, $\mathbf{P}^{\{i\}}$, $\mathbf{B}^{\{i\}}$, iteration number Ξ.
Ensure: $\mathbf{P}^{\{i+1\}}$.
Init: feasible point $\mathbf{P}^{[0]} \leftarrow \mathbf{P}^{\{i\}}$.
$\mathbf{F}^* \leftarrow \mathbf{F}^{\{i+1\}}$, $\mathbf{B}^* \leftarrow \mathbf{B}^{\{i\}}$.
for $\xi = 1$ to Ξ **do**
  $\widehat{\mathbf{P}} \leftarrow \mathbf{P}^{[\xi-1]}$.
  Construct the approximation function $\underline{R}^{\mathrm{P}}_{s,k}(\mathbf{P};\widehat{\mathbf{P}})$ according to (36).
  Solve (P2-ξ) and obtain $\mathbf{P}^{[\xi]}$.
end for
$\mathbf{P}^{\{i+1\}} \leftarrow \mathbf{P}^{[\Xi]}$.

C. Quadratic Programming for Load Balancing

After determining BH pattern $\mathbf{F}^*$ and resource management $\mathbf{P}^*$, the subproblem for load balancing $\mathbf{B}$ is established as

$$(\text{P3}): \min_{\mathbf{B}} \ \Lambda(\mathbf{F}^*, \mathbf{P}^*, \mathbf{B}; V) \tag{38}$$
$$\text{s.t.} \quad (13\text{h}), (13\text{i}), (13\text{j}), \text{ and } (13\text{l}),$$

where $d_{s,k}$ is linear with $\mathbf{B}$ as indicated in (6), and thus the quadratic term in the objective is convex. Besides, the constraints (13h), (13i), and (13l) are also linear with $\mathbf{B}$. However, for the amount of bidirectionally transferred packets $c_{r,s}$ through the laser ISL, the absolute value terms of $\mathbf{B}$ are included in (5). Therefore, auxiliary variables $b^+_{r,s,k}$ are introduced, and the following constraints are imposed as

$$b^+_{r,s,k} \geq b_{r,s,k}, \tag{39a}$$
$$b^+_{r,s,k} \geq -b_{r,s,k}, \tag{39b}$$
$$c_{r,s} = \sum_{k\in\mathcal{K}} b^+_{r,s,k}. \tag{39c}$$

Since the objective function and constraint (13j) tend to reduce $c_{r,s}$, exactly one of the inequalities between constraints (39a) and (39b) holds with equality in the final solution, which ensures the equivalence of the solution after incorporating these constraints. Therefore, the objective function of this subproblem is a quadratic convex problem, while all constraints can be transformed into linear constraints, allowing the problem to be solved via quadratic programming.

D. Convergence Analysis and Computational Complexity

For the convergence of each sub-problem, the MPMM method for the problem (P1) has been proved to be convergent in [41] and [42]. Moreover, in terms of the problem (P2), since $\underline{R}^{\mathrm{P}}_{s,k}(\mathbf{P};\mathbf{P}^{[\xi]})$ serves as a lower bound of $R_{s,k}(\mathbf{P})$, the optimal solution of (P2-ξ) is feasible for (P2), and the objective function is ensured to be non-increasing at each iteration. Therefore, the result sequence $\{\Lambda(\mathbf{F}^*, \mathbf{P}^{[\xi]}, \mathbf{B}^*; V) | \xi = 1, 2, \ldots\}$ is monotone and bounded, ensuring its convergence. Additionally, the optimal solution to the problem (P3) can be obtained directly via quadratic programming.

Therefore, for the BCD algorithm, its objective value is non-increasing in each optimization step as

$$\begin{aligned}&\Lambda(\mathbf{F}^{\{i\}},\mathbf{P}^{\{i\}},\mathbf{B}^{\{i\}};V) \leq \Lambda(\mathbf{F}^{\{i+1\}},\mathbf{P}^{\{i\}},\mathbf{B}^{\{i\}};V)\\ &\leq \Lambda(\mathbf{F}^{\{i+1\}},\mathbf{P}^{\{i+1\}},\mathbf{B}^{\{i\}};V) \leq \Lambda(\mathbf{F}^{\{i+1\}},\mathbf{P}^{\{i+1\}},\mathbf{B}^{\{i+1\}};V).\end{aligned} \tag{40}$$

Additionally, because the transmission power is constrained by $P_{\max}$ and the queue lengths are non-negative, the objective function has a lower bound. Consequently, the sequence of objective values generated by each optimization step is monotonically non-increasing and bounded, ensuring the convergence of the BCD algorithm.

For the computational complexity, in terms of **Algorithm** 2, given that the accuracy of the solution is ϵ_1, the complexity of solving convex problem (P1-θ-ψ) with the interior-point method (IPM) is $\mathcal{O}\left((|\mathcal{S}||\mathcal{K}|)^{3.5}\log(1/\epsilon_1)\right)$ [43]. Moreover, both the MP and MM algorithms exhibit linear convergence [41], [42], and thus the round and iteration numbers are $\Theta = \mathcal{O}(\log(1/\epsilon_1))$, $\Psi = \mathcal{O}(\log(1/\epsilon_1))$. Therefore, the computational complexity ω_1 of **Algorithm** 2 can be expressed as $\omega_1 = \mathcal{O}\left((|\mathcal{S}||\mathcal{K}|)^{3.5}(\log(1/\epsilon_1))^3\right)$.

Given the precision ϵ_2 for **Algorithm** 3, the iteration number $\Xi = \mathcal{O}(\log(1/\epsilon_2))$, and the complexity of solving convex optimization problem (P2-ξ) by IPM is $\mathcal{O}\left((|\mathcal{S}||\mathcal{K}||\mathcal{L}|)^{3.5}\log(1/\epsilon_2)\right)$. Consequently, the computational complexity is $\omega_2 = \mathcal{O}\left((|\mathcal{S}||\mathcal{K}||\mathcal{L}|)^{3.5}(\log(1/\epsilon_2))^2\right)$.

Moreover, the computational complexity of quadratic programming for load balancing can be calculated as $\omega_3 = \mathcal{O}\left((|\mathcal{S}|^2|\mathcal{K}|)^{3.5}\log(1/\epsilon_3)\right)$ with the precision ϵ_3.

In summary, each loop of the BCD algorithm in **Algorithm** 1, encompassing the implementation of the three aforementioned algorithms, can be completed in polynomial time. Moreover, the loop limit is set as Π, and the total computational complexity of the proposed method is $\omega_0 = \mathcal{O}(\Pi(\omega_1 + \omega_2 + \omega_3))$.

V. Simulation Results

The simulation parameters and the implementation of the proposed method are discussed in Section V-A. Additionally, in Section V-B, the convergence results of the proposed method are illustrated, along with its performance under different parameters. Besides, comparisons with other benchmarks are provided in Section V-C.

A. Simulation Parameters and Implementation

The simulation parameters are summarized in Table II. A Walker constellation with an orbital altitude of 1,000 km and an inclination of 85° is employed, where the distance between two adjacent satellites is set to 207 km to align with the cell coverage configuration. The specifications of the laser ISLs are based on the CONDOR laser terminal developed by Mynaric [40]. The time slot duration is set to 10 ms [12], and each simulation consists of 1,500 time slots, corresponding to a total duration of 15 seconds. Due to the short simulation period, the satellite-to-cell coverage relationships, i.e., $\Omega(s)$ and $\Phi(k)$, are assumed to remain constant [20], [37], [44], while the laser ISL performance is also considered stable,

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TABLE II
SIMULATION PARAMETERS

Parameters	Value
The number of satellites, $\lvert\mathcal{S}\rvert$	9
The number of cells per satellite, $\lvert\Omega(s)\rvert$	19
The total number of cells, $\lvert\mathcal{K}\rvert$	99
The number of beams per satellite, N_b	4
The radius of each cell	39 km
The satellite altitude	1,000 km
The carrier frequency	20 GHz
The bandwidth of each frequency segment, W	50 MHz
The number of frequency segment, $\lvert\mathcal{L}\rvert$	4
The transmitter antenna gain	32.44 dBi
The receiver antenna gain	26.42 dBi
The noise spectral power density, N_0	-204 dBW/Hz
The interference threshold, Z_{th}	-130 dBW
The achievable data rate of laser ISL, R_{I}	10 Gbps
The fixed power of laser ISL, P_{I}	4 W
The duration of each time slot	10 ms
The fixed size of data packet, M_0	5 Kbit
The normalized factors, L_0, E_0	10^6, 6.25 J

as the inter-satellite distance changes insignificantly. For data transmission, the carrier frequency is set to 20 GHz in the Ka-band [20], while the radiation patterns of the transmitter and receiver antennas follow ITU-R S.1528 [45] and ITU-R S.465-6 [46], respectively. Accordingly, the channel coefficients are calculated based on models in [12], [15], and [16], and are updated in each time slot. For other channel models, the proposed method remains applicable as long as the channel coefficients can be modeled as Markov processes, while its performance is still ensured by *Proposition* 1, as demonstrated in Appendix A. In addition, the average packet arrival rate for each cell is uniformly distributed between 80 Mbps and 320 Mbps, with a gradual drift occurring every 50 time slots. Moreover, the number of arriving packets follows a Poisson distribution [21].

For the implementation of the proposed method, a centralized architecture is adopted, where a designated node is responsible for multi-satellite synchronization and centralized information collection, including the current buffer queue lengths, channel state information, and spectrum overlapping status with GSO ground stations from all satellites. Given that the BH pattern selection, resource management, and load balancing have polynomial time computational complexity, the proposed method achieves high computational efficiency when implemented on an application-specific integrated circuit [47]. Moreover, due to the extremely high transmission rate of laser ISLs [40], the dominant portion of algorithm execution time is attributed to the propagation delay in inter-satellite communication. Meanwhile, in the simulation, the distance between two adjacent satellites is set as 207 km to align with the cell coverage configuration, resulting in a round-trip propagation delay per laser ISL of approximately 1.38 ms. For small-scale constellations, such latency allows the satellites to implement information exchange within each time slot, ensuring the feasibility of multi-satellite collaboration.

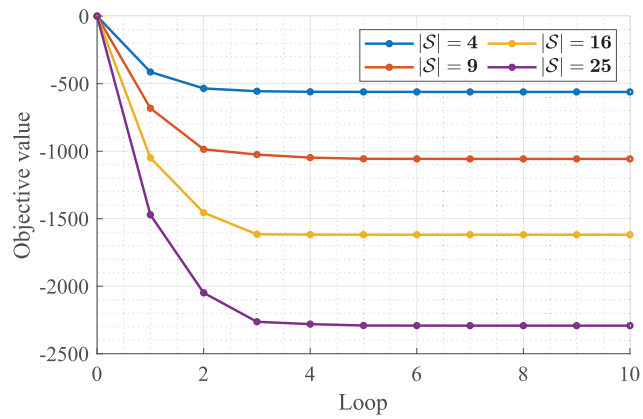

Fig. 2. The iteration curves of the BCD algorithm in various satellite numbers.

For larger-scale constellations, the system can be partitioned into multiple regions, with each performing coordination independently.

With the ongoing advancement of non-terrestrial networks and satellite communication standards, the proposed method is expected to facilitate the integration of multi-satellite collaboration into mainstream standards such as 3GPP, thereby accelerating its practical deployment in LEO satellite-terrestrial communication systems. In addition, the subsequent simulation results demonstrate that the proposed multi-satellite collaboration approach achieves superior performance under typical scenarios and parameter settings, highlighting its practical value and application potential.

B. Performance in Different Parameters

In Fig. 2, the iteration curves of the objective function value $\Lambda(\mathbf{F}, \mathbf{P}, \mathbf{B}; V)$ obtained at the end of each loop in the BCD algorithm for (P0-2) with varying numbers of LEO satellites are presented. To ensure a fair comparison, the average queue length at each satellite is uniformly distributed between 0 and 4,000, while the trade-off coefficient V is set to 200. It is observed that the objective function decreases rapidly during the initial loop and stabilizes after about four loops. These results indicate that the proposed BCD algorithm is always effective regardless of the number of satellites.

In addition, the average queue length and energy consumption performance of LEO satellite communications for different trade-off coefficients V are illustrated in Fig. 3, with communication demand of 20 Gbps, and the results are averaged over 30 time slots. As shown in Fig. 3(a), the average queue length increases proportionally with V. Moreover, for each value of V, the queue length eventually stabilizes, indicating that the service capacity of LEO satellites is sufficient to meet the communication demands, thereby satisfying constraint (13a). Meanwhile, in Fig. 3(b), the energy performance is represented by the power E_s/T_0 per time slot. It can be observed that, as V increases, the energy consumption decreases gradually, though at a diminishing rate, which reflects the trade-off of $\mathcal{O}(V) \sim \mathcal{O}(1/V)$ between the average queue length and energy consumption, as outlined in *Proposition* 1. Moreover, the queue length is proportional to

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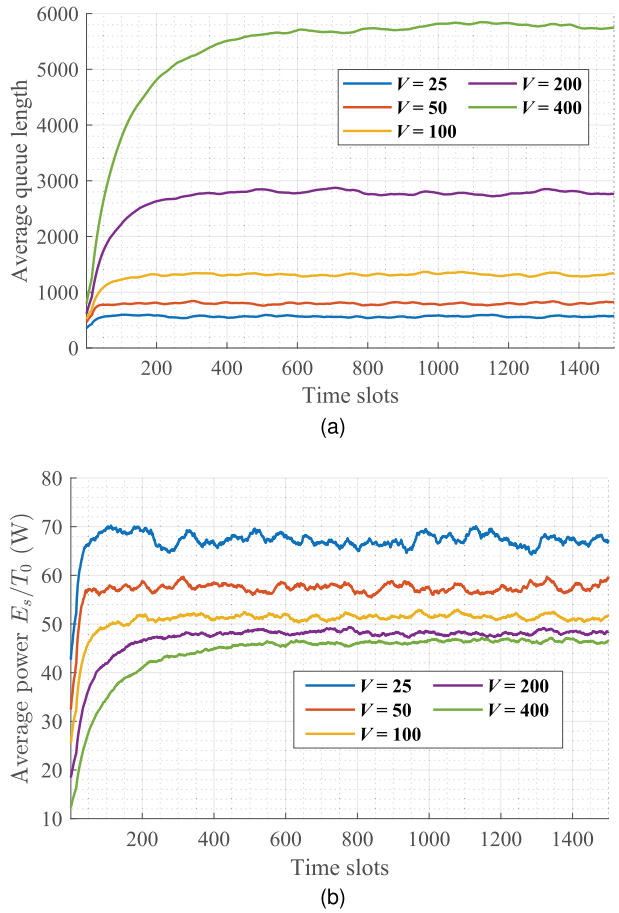

Fig. 3. The performance per time slot versus (a) the average queue length and (b) the average power when the communication demand is 20 Gbps.

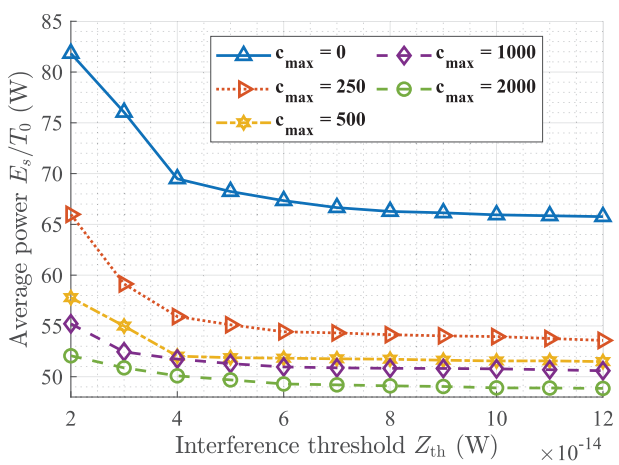

Fig. 4. The average power for various interference thresholds under different laser ISL transfer limits.

the average queuing latency, which implies that reducing the energy consumption comes at the cost of increased queuing latency. Therefore, selecting an appropriate trade-off coefficient is crucial. Since the case of $V = 200$ yields lower energy consumption while maintaining a moderate queue length, it is applied in subsequent simulations.

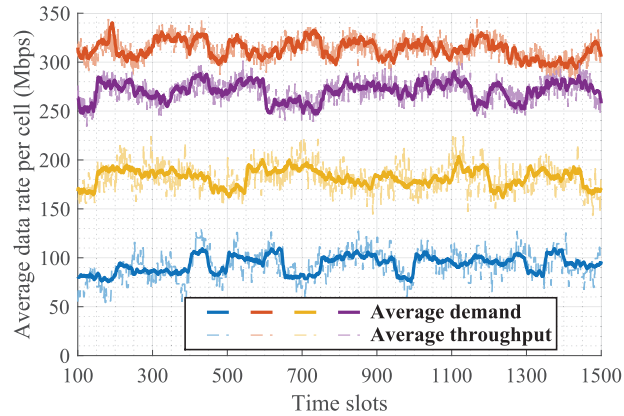

Fig. 5. The average demands and throughputs for four representative cells.

Moreover, in Fig. 4, the energy consumption required to satisfy a communication demand of 20 Gbps is presented under varying interference thresholds at the GSO ground station with different laser ISL transfer limits. When the interference threshold is strictly constrained, higher energy consumption is required due to limited available spectrum resources. As the interference threshold is gradually higher, the energy consumption correspondingly decreases. It is observed that increasing the laser ISL transfer limit can also reduce the energy consumption with various interference thresholds. This is because satellites severely affected by interference constraints tend to accumulate longer queues, while the proposed real-time load balancing scheme transfers packets to nearby satellites with lower interference based on the queue lengths. Hence, the proposed method effectively suppresses the interference in co-existing systems. Moreover, with more packets allowed for laser ISL transfer, lower energy consumption is achieved, which is attributed to improved handling of non-uniform distribution of communication demands by real-time load balancing, avoiding overloading any single satellite and thereby reducing the average energy consumption of LEO satellite communications.

Meanwhile, in Fig. 5, four representative cells are selected, including those with the lowest and highest average demands and two others with moderate demand levels, where their average demands and average throughputs are depicted. Since initial queue lengths are set to zero, several beginning time slots are required to reach steady state, and only the steady-state results are presented. It is observed that despite significant differences and temporal fluctuations in average demands among the cells, the average throughput in each cell closely follows its demand. This demonstrates that the proposed method fairly satisfies the service requirements across all cells. Consequently, the fairness among cells is achieved.

C. Comparison With Other Benchmarks

In this paper, the performance of the following benchmarks are compared with the proposed method: 1) **DRL for BH pattern** [21], which employs DRL to determine the BH pattern, followed by resource management and load balancing based on the selected pattern. Specifically, the state of the DRL

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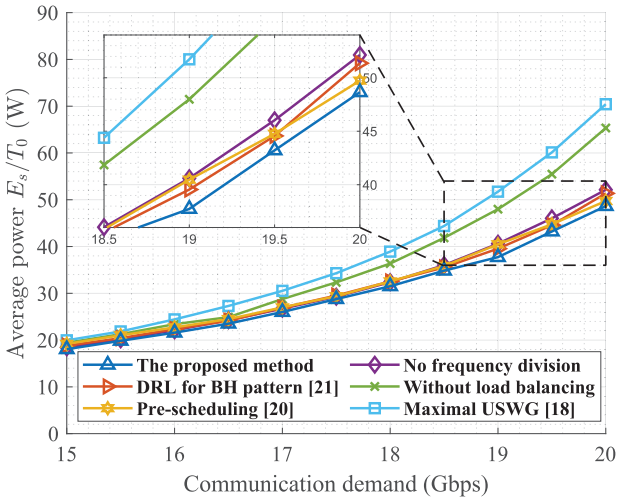

Fig. 6. The average power of different methods on various communication demands with unlimited storage volume.

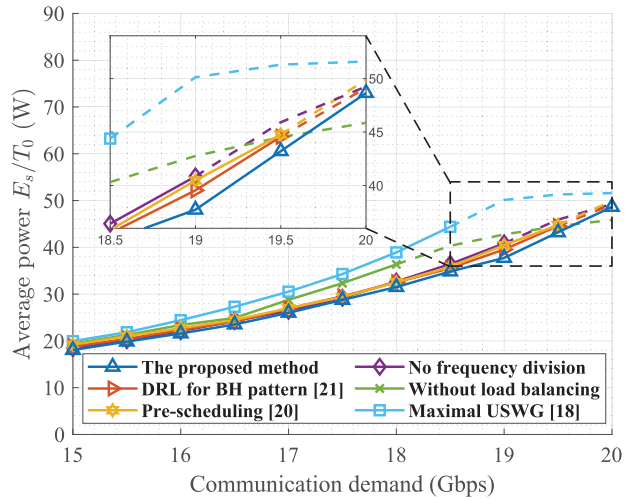

Fig. 7. The average power of different methods on various communication demands with limited storage volume.

includes channel coefficients $h_{s,k,j}[n]$, $g_{s,k,j}[n]$, and queue length $\mathbf{Q}[n]$, while the reward in each time slot is defined by (19). Additionally, the QMIX architecture with centralized training and distributed execution is utilized [21]. 2) **Pre-scheduling** [20], where the total volume of the data packets arriving in each cell among the future 50 time slots is estimated via traffic prediction, based on which the BH pattern, resource management, and load balancing are scheduled in advance. 3) **No frequency division**, where the power of frequency segments overlapping with the GSO ground station is set to zero, while equal power is assigned to the remaining frequency segments in each beam, but the power can be various among different beams. 4) **Without load balancing**, where the load balancing component is removed from the proposed method. 5) **Maximal USWG** [18], where the weight of each cell is determined by its queue length and latency, with higher-weighted cells prioritized for service.

The average energy consumption of various approaches under different communication demands is illustrated in Fig. 6, with the fixed trade-off coefficient $V = 200$. Moreover, it is assumed that each satellite has sufficient storage volume, i.e., no packet loss occurs. Among all methods, the proposed approach consistently exhibits the lowest average power consumption, whereas the DRL for BH pattern method performs slightly worse due to the challenges of adapting to time-varying characteristics and insufficient information during distributed execution. In addition, DRL requires substantial computational resources and training samples during offline learning. Meanwhile, it can be observed that as communication demand increases, the power consumption of the pre-scheduling scheme gradually approaches that of the proposed method, since higher communication demand leads to increased queue lengths, reducing the impact of traffic arrival randomness. Besides, no frequency division method results in inferior performance due to the reduced flexibility for resource management. For the benchmark without load balancing, some satellites become overloaded as the communication demand increases, leading to a significant rise in energy consumption, while the maximal USWG scheme fails to account for the time-varying characteristics, resulting in deficient performance. Particularly, when the communication demand is 20 Gbps, the proposed method reduces the average power by 5.28%, 2.17%, 6.68%, 25.54%, and 30.92%, compared with DRL for BH pattern, pre-scheduling, no frequency division, without load balancing, and maximal USWG, respectively.

Meanwhile, in Fig. 7, the storage volume of each satellite is limited to 350 Mbit, and the excess packets are dropped, where the dashed lines in the curves indicate that packet loss occurs. The proposed method can satisfy all communication demands within the range of $15 \sim 20$ Gbps, demonstrating its effectiveness in maintaining the queue stability. Nevertheless, the DRL for BH pattern method begins experiencing packet loss at 19.5 Gbps, while the pre-scheduling scheme also experiences packet loss at this point, due to its neglect of the randomness in traffic arrival. Additionally, packet loss appears earlier in other methods, and although the method without load balancing has less energy consumption than the maximal USWG scheme, the packet loss occurs earlier since some satellites are overloaded.

In addition, the average power consumption and queue length of all algorithms are illustrated in Fig. 8 under a communication demand of 17.5 Gbps, at which all methods are capable of fully satisfying the communication demand, and the results are obtained by varying the trade-off coefficient V. It can be seen that increases in energy consumption yield minimal reductions in queue length once the energy consumption becomes sufficiently high, and the queue length sharply increases as the energy consumption reduces to a certain threshold, clearly demonstrating the trade-off between queueing latency and energy consumption proposed in *Proposition* 1, and underscoring the importance of selecting an appropriate trade-off coefficient V. Moreover, the proposed method consistently achieves both the lowest energy consumption and the shortest queue length under the same conditions. While the DRL for BH pattern method shows comparable performance under moderate conditions, e.g., average power

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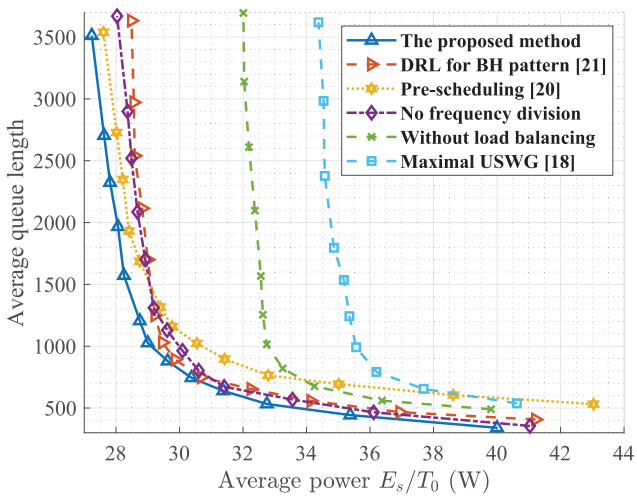

Fig. 8. The average power and queue length curves under different methods.

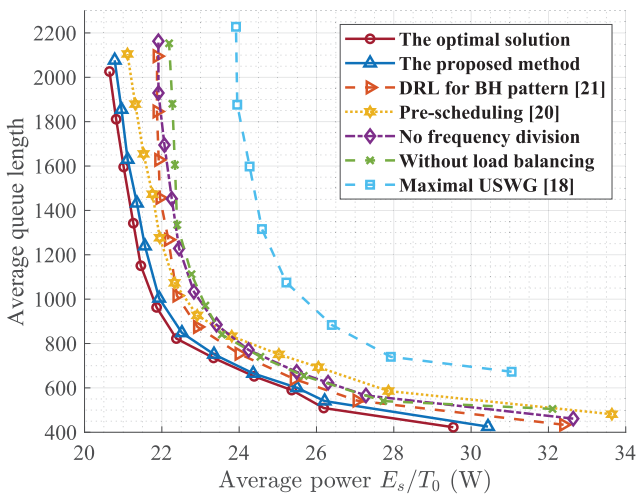

Fig. 9. The average power and queue length curves in a small-scale scenario under the exhaustive-search-based optimal solution and other methods.

around 30 W, but deteriorates in other scenarios. For the pre-scheduling scheme, as the queue length increases, its performance approaches that of the proposed method, consistent with previous findings. In contrast, the no frequency division method consistently underperforms the proposed approach, and both the method without load balancing and the maximal USWG scheme exhibit inferior performance. Therefore, the superiority of the proposed method has been demonstrated in terms of reducing the energy consumption and satisfying the communication demands of LEO satellites.

Additionally, to illustrate the gap between the proposed method and the possible optimal solution, a small-scale scenario is simulated in Fig. 9, involving 2 satellites, each equipped with 2 beams, covering 12 terrestrial cells, with the communication demand set at 2.5 Gbps. The trade-off between the average power and average queue length for various methods is evaluated. Specifically, the optimal solution is obtained through an exhaustive search for comparison with other algorithms. It can be observed that the proposed method remains closest to the optimal solution across all the benchmarks. Additionally, due to the limited number of satellites and coverage cells in this small-scale scenario, the negative impact of the benchmark without load balancing is less pronounced than that in Fig. 8. Moreover, the trends of the remaining methods are similar to those observed in Fig. 8 and inferior to the proposed method.

VI. Conclusion

In this paper, the long-term average energy minimization problem in LEO satellite communications was addressed. By jointly considering the BH pattern, resource management, and load balancing, the proposed method guaranteed the queue stability and interference suppression while significantly reducing the long-term average energy consumption. Specifically, the Lyapunov drift-plus-penalty method transformed the long-term problem into optimization within each time slot. Moreover, the BCD algorithm was employed to alternately optimize different variables. In particular, the BH pattern was obtained via the MPMM approach, while the non-convex constraints in the subproblem of resource management were handled by the SCA technique, and real-time load balancing was realized through quadratic programming. Simulation results demonstrated that the proposed method substantially reduced the energy consumption while maintaining the queue stability and ensuring that the interference to co-existing systems remained below the prescribed threshold, outperforming other benchmarks. In conclusion, the proposed method offers a valuable view for LEO satellite communications with limited SWaP.

Appendix A
Proof of Proposition 1

In a policy that depends only on the current channel coefficients, the variables are denoted as $\widehat{a}_{s,k}$, $\widehat{d}_{s,k}$, $\widehat{x}_{s,k}$, and $\widehat{E}_s$, and then for all these policies, including the optimal policy, the following inequality for one-slot drift is given by

$$\tilde{\mathfrak{L}}[\nu] = \frac{L_{\mathrm{q}}[\nu+1]-L_{\mathrm{q}}[\nu]}{L_0} + \frac{V}{E_0}\sum_{s\in\mathcal{S}} E_s[\nu] \tag{41}$$

$$\leq \frac{\Upsilon_1}{L_0} + \frac{1}{L_0}\sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}\Big(\frac{1}{2}d_{s,k}^2[\nu] + q_{s,k}[\nu](a_{s,k}[\nu]+d_{s,k}[\nu] - x_{s,k}[\nu])\Big) + \frac{V}{E_0}\sum_{s\in\mathcal{S}} E_s[\nu] \tag{41a}$$

$$\leq \frac{\Upsilon_1}{L_0} + \frac{1}{L_0}\sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}\Big(\frac{1}{2}\widehat{d}_{s,k}^2[\nu] + q_{s,k}[\nu](\widehat{a}_{s,k}[\nu]+\widehat{d}_{s,k}[\nu] - \widehat{x}_{s,k}[\nu])\Big) + \frac{V}{E_0}\sum_{s\in\mathcal{S}} \widehat{E}_s[\nu] \tag{41b}$$

$$\leq \frac{1}{L_0}\left((2(\nu-n_0)+1)\Upsilon_1 + \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}(\nu-n_0)d_{\max}^2\right) + \frac{1}{L_0}\sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}\Big(\frac{1}{2}\widehat{d}_{s,k}^2[\nu] + q_{s,k}[n_0](\widehat{a}_{s,k}[\nu]+\widehat{d}_{s,k}[\nu] - \widehat{x}_{s,k}[\nu])\Big) + \frac{V}{E_0}\sum_{s\in\mathcal{S}} \widehat{E}_s[\nu], \tag{41c}$$

where $\nu \geq n_0$, and (41a) holds according to (18). Moreover, due to the result obtained by the problem (P0-2) aims to

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minimize (41a), the inequality (41b) is true for all other policies, while (41c) holds due to the following formula as

$$q_{s,k}[\nu] = q_{s,k}[n_0] + \sum_{n=n_0}^{\nu-1} (a_{s,k}[n] + d_{s,k}[n] - x_{s,k}[n]). \quad (42)$$

By substituting it into (41b) and scaling the result to its upper bound, the inequality (41c) can be derived.

As indicated in [38], the channel coefficients in satellite-terrestrial communications can be modeled as a Markov process. It is assumed that at the n_0-th time slot, the channel is in initial state, and n_r denotes the time slot index at which the channel returns to the initial state for the r-th time. In addition, the recurrence time, expressed in terms of the number of time slots, is defined as the random variable u. Therefore, by summing $\tilde{\mathfrak{L}}[\nu]$ from the n_r-th time slot to the $(n_{r+1}-1)$-th time slot, the inequality for multi-slot drift $\mathfrak{L}_r$ is obtained as

$$\begin{aligned}\mathfrak{L}_r &= \mathbb{E}\left\{\frac{L_{\mathrm{q}}[n_{r+1}] - L_{\mathrm{q}}[n_r]}{L_0} + \frac{V}{E_0}\sum_{n=n_r}^{n_{r+1}-1}\sum_{s\in\mathcal{S}} E_s[n]\middle|\mathbf{Q}[n_r]\right\} \\ &\le \frac{\Upsilon_2}{L_0} + \mathbb{E}\{u\}\Big(\frac{1}{L_0}\sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}} q_{s,k}[n_r]\mathbb{E}\{\widehat{a}_{s,k}[n] \\ &\quad + \widehat{d}_{s,k}[n] - \widehat{x}_{s,k}[n]\} + \frac{V}{E_0}\sum_{s\in\mathcal{S}}\mathbb{E}\{\widehat{E}_s[n]\}\Big), \end{aligned} \quad (43)$$

where Υ_2 is a constant term defined as

$$\begin{aligned}\Upsilon_2 &= \mathbb{E}\left\{u^2\right\}\left(\Upsilon_1 + \sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}\frac{d_{\max}^2}{2}\right) \\ &\quad + \mathbb{E}\{u\}\mathbb{E}\left\{\sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}}\frac{d_{s,k}^2[n] - d_{\max}^2}{2}\right\}. \end{aligned} \quad (44)$$

Inequality (43) holds because in an irreducible recurrent Markov chain, each recurrence time is independently and identically distributed, and the policy depends only on the current channel coefficients. Moreover, the optimal policy has the following properties as

$$\mathbb{E}\{\widehat{a}_{s,k}[n] + \widehat{d}_{s,k}[n] - \widehat{x}_{s,k}[n]\} \le 0, \quad (45a)$$

$$\sum_{s\in\mathcal{S}}\mathbb{E}\{\widehat{E}_s[n]\} = \mathcal{E}_{\mathrm{opt}}. \quad (45b)$$

Substituting (45a) and (45b) into (43), we have an upper bound of multi-slot drift $\mathfrak{L}_r$ as

$$\mathfrak{L}_r \le \frac{\Upsilon_2}{L_0} + \frac{V}{E_0}\mathbb{E}\{u\}\mathcal{E}_{\mathrm{opt}}. \quad (46)$$

Considering the duration from the n_0-th to $n_{\mathcal{R}}$-th time slot, which corresponds to the $\mathcal{R}$-th return to the initial state, the above formula (46) is accumulated from the n_0-th to $(n_{\mathcal{R}}-1)$-th time slot, yielding the $\mathcal{R}$-return drift as

$$\sum_{r=0}^{\mathcal{R}-1}\mathfrak{L}_r = \frac{\mathbb{E}\{L_{\mathrm{q}}[n_{\mathcal{R}}]\} - \mathbb{E}\{L_{\mathrm{q}}[n_0]\}}{L_0} + \frac{V}{E_0}\mathcal{R}\mathbb{E}\{u\}\mathcal{E} \quad (47a)$$

$$\le \frac{\mathcal{R}\Upsilon_2}{L_0} + \frac{V}{E_0}\mathcal{R}\mathbb{E}\{u\}\mathcal{E}_{\mathrm{opt}}. \quad (47b)$$

Dividing both (47a) and (47b) by $\mathcal{R}$, and letting $\mathcal{R}$ approach infinity yields

$$\mathcal{E} \le \mathcal{E}_{\mathrm{opt}} + \frac{E_0\Upsilon_2}{VL_0\mathbb{E}\{u\}}, \quad (48)$$

thereby verifying (21).

For the average queue length, when the Slater condition holds, a channel coefficient based policy can satisfy that

$$\mathbb{E}\{\widehat{a}_{s,k}[n] + \widehat{d}_{s,k}[n] - \widehat{x}_{s,k}[n]\} \le -\delta < 0, \quad (49a)$$

$$\sum_{s\in\mathcal{S}}\mathbb{E}\{\widehat{E}_s[n]\} = \widehat{\mathcal{E}}. \quad (49b)$$

Similar to (46) and (47), by calculating the $\mathcal{R}$-return drift and taking the limit, the average queue length can be derived as

$$\lim_{|\mathcal{N}|\to\infty}\frac{1}{|\mathcal{N}|}\mathbb{E}\left\{\sum_{n\in\mathcal{N}}\sum_{s\in\mathcal{S}}\sum_{k\in\mathcal{K}} q_{s,k}[n]\right\} \le \frac{1}{\delta}\left(\frac{\Upsilon_2}{\mathbb{E}\{u\}} + \frac{VL_0\widehat{\mathcal{E}}}{E_0}\right), \quad (50)$$

thereby approving (22). □

Appendix B
Proof of Proposition 2

Consider the logarithmic function $w_1(\gamma) = \log_2(1+\gamma)$. According to [48], a lower bound for $w_1(\gamma)$ is defined as

$$\underline{w}_1(\gamma) = \frac{\hat{\gamma}}{1+\hat{\gamma}}\log_2(\gamma) + \log_2(1+\hat{\gamma}) - \frac{\hat{\gamma}}{1+\hat{\gamma}}\log_2(\hat{\gamma}). \quad (51)$$

This bound satisfies the conditions as $\underline{w}_1(\hat{\gamma}) = w_1(\hat{\gamma})$, $\frac{d\underline{w}_1}{d\gamma}|_{\gamma=\hat{\gamma}} = \frac{dw_1}{d\gamma}|_{\gamma=\hat{\gamma}}$. Substituting (51) into $R_{s,k}(\mathbf{F})$ leads to the inequality as

$$R_{s,k}(\mathbf{F}) \ge \sum_{l\in\mathcal{L}} W f_{s,k}\left(\zeta_{s,k,l}^{\mathrm{F}}(\widehat{\mathbf{F}}) - \phi_{s,k,l}^{\mathrm{F}}(\widehat{\mathbf{F}})\log_2\left(\frac{1}{\gamma_{s,k,l}(\mathbf{F})}\right)\right). \quad (52)$$

Focusing on the first term in (52), a new function $w_2(\gamma) = \log_2(1/\gamma)$ is defined, which admits an upper bound as

$$\overline{w}_2(\gamma) = \frac{\hat{\gamma}\log_2(e)}{\gamma} - \log_2(e\hat{\gamma}). \quad (53)$$

Similar to $\underline{w}_1(\gamma)$, it holds that $\underline{w}_2(\hat{\gamma}) = w_2(\hat{\gamma})$, $\frac{d\underline{w}_2}{d\gamma}|_{\gamma=\hat{\gamma}} = \frac{dw_2}{d\gamma}|_{\gamma=\hat{\gamma}}$. Therefore, the following inequality is derived as

$$\begin{aligned}R_{s,k}(\mathbf{F}) \ge \sum_{l\in\mathcal{L}} W\Big(&\big(\phi_{s,k,l}^{\mathrm{F}}(\widehat{\mathbf{F}})\log_2\left(e\gamma_{s,k,l}(\widehat{\mathbf{F}})\right) \\ &+ \zeta_{s,k,l}^{\mathrm{F}}(\widehat{\mathbf{F}})\big)f_{s,k} \\ &- \log_2(e)\phi_{s,k,l}^{\mathrm{F}}(\widehat{\mathbf{F}})\gamma_{s,k,l}(\widehat{\mathbf{F}})\frac{N_0W + I_{s,k,l}(\mathbf{F})}{|h_{s,k,k}|^2 p_{s,k,l}^*}\Big). \end{aligned} \quad (54)$$

Since the right term is linear in $\mathbf{F}$, a regularization term $-\kappa(f_{s,k} - \hat{f}_{s,k})^2$ is added in (31) to ensure the strong concavity. By applying variable substitutions into (52) and (54) along with the chain rule, the *Proposition* 2 is confirmed. □

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